Fiscal Year:	FY 2011	**Task Last Updated:**	FY 08/22/2010
PI Name:	Grabham, Peter Ph.D.		
Project Title:	Development and use of human 3-Dimensional tissue culture models for the study of space radiation effects on the central nervous system		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	pwg2@cumc.columbia.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	646-761-1275
Organization Name:	Columbia University		
PI Address 1:	Center for Radiological Research		
PI Address 2:	630W W 168th St, VC 11-243		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10032-3702	**Congressional District:**	13
Comments:			
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
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Task Description:	Human space exploration involves exposure to types of radiation not normally encountered on earth. As well as carcinogenesis, the effect of these particles on the human central nervous system is of particular concern since impairment of cognitive function could not only compromise missions but cause later effects on the quality of life. Studies in animals have shown that space radiation particles can have adverse effects on cognition and behavior. The effect on neurogenesis has also been documented and proposed to be a cause of cognitive loss. Information regarding the effects of space radiation on the CNS is however, still in its infancy due to the difficulty of extrapolating data from animals and cell lines to the intact human being. The primary objective of the research in this proposal is to increase our understanding of the effects of space radiation on the human central nervous system with an emphasis on neurodegeneration. We are using a relatively new approach to this question by using available human stem cells to create 3-dimensional tissue models. These can represent human neural tissue more closely than 2-dimensional monolayers. Human cells of three of the main types in the brain will be cultured and differentiated in 3-D matrix gels and used to determine the effects of space radiation by assaying for specific endpoints related to degeneration in neurons. This is achieved by live and fixed stain 3-D imaging together with assays for secreted proteins. 3-D cultures are grown in gels as monocultures and as a multi-cell 'neurovascular unit' built on an existing model of human vessels. Such a model can complement animal and cell studies to produce new and more complete data on the effects of space radiation on the human nervous system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of 3-Dimensional human tissue models from normal human cells and stem cells is a relatively new technology but one with great potential in many fields of medical research. Tissue models can more accurately depict human tissue since the cells can be arranged spatially as they would be in vivo and can interact with each other as they would in the human body. A neurovascular unit can be used for basic research on many aspects of the human brain. These include regeneration, synaptic function, and degeneration. Because the tissue model is derived from individual cells, each cell type can be altered genetically before it is incorporated into the model. Such modification has a great advantage in that specific aspects of brain function can be targeted.
Task Progress:	We have attempted to approach the development of the human brain tissue model in a methodical and structured way. First, we acquired at least 3 human cell types for each of the components of the model; neurons, astrocytes and endothelial cells, to ensure the best possible model and provide back-up cells. Second, stem cells are the most suitable source of neurons since they are closest to normal, so we have developed our own multistep protocol for the generation and differentiation of normal human neurons from these cells and successfully detected expression of neuronal markers and degeneration markers in 3D monocultures. Astrocytes and neuroblastoma are differentiated by treatment with retinoic acid and endothelial cells can be readily induced to form microcapillaries in 3-D matrices. The strategy for constructing tissue models is to embed neurons and astrocytes into protein gel matrices after they have been differentiated into the respective cell type then induce morphological or 'endstage' differentiation while they are in the matrix. Endothelial cells can then be added later, given the right factors, and allowed to grow into the matrix and form blood vessels as in vivo during angiogenesis. We have tested different media types and different pairings of cell types and found that the requirement of endothelial cells is not entirely compatible with that of neurons and astrocytes. Endothelial cells, which line the vessels and are in contact with blood in vivo require serum for vessel formation. Neurons and Astrocytes, normally behind the blood brain barrier (BBB), require serum-free conditions to differentiate in matrices. However, we were able to create a prototype neurovascular unit using the above protocol by reducing the serum concentration and relying on trophic support from astrocytes for the maintenance of vessels. We are currently improving the longevity of these models and substituting neurons from stem cells for the neuroblastoma used in our prototype model. We have begun to develop the assays for degeneration before the 'neurovascular unit' is completed. Some assays need time to develop and the single cell type and paired cell type 3D cultures also serve as controls for the model in addition to providing important data in their own right. We have started to assay the effects of space radiation on the morphology of neuron/astrocyte 3D models. Results are to date, preliminary, and further experiments will confirm the effect of these radiations on neurons in the presence of astrocytes. Immunocytochemistry has enabled us to detect the amyloid precursor protein (APP) that is associated with Alzheimers disease, intracellularly and at the cell surface. Furthermore, we can also detect, the cleaved products of APP that are released into the media, thus we are able to track the expression, localization and cleavage of this degeneration marker after irradiation and can therefore monitor some of the events that occur during Alzheimers degeneration. SH-SY-5Y neuroblastoma cells, which can be easily differentiated, have a particular advantage in that they over express APP. Correct functioning of the blood-brain-barrier is important in brain function and compromise of this barrier between the blood supply and the brain cells is thought to be a cause of a number of pathologies. When the barrier is more permeable blood proteins come into contact with neurons and cause damage. We are developing an assay that does not utilize the neurovascular unit but still includes a cell mixture of endothelial cells and astrocytes. Thus it will represent the human blood-brain-barrier more closely than endothelial cells alone. Endothelial cells and astrocytes can be grown on a permeable membrane of the inner chamber in transwell dishes, where they form a barrier to protein diffusion just like in the brain. We have made trans-endothelial electrical resistance (TEER) measurements quickly and non-invasively by inserting electrodes into each chamber. Preliminary data shows that endothelial cells plus astrocytes give elevated TEER measurements and thus form a barrier to permeability just like in the brain. However, experiments at NSRL with this protocol proved to be technically difficult, the TEER measurements were not reliable. We have therefore moved on to an assay for protein diffusion across cell layers since this is the endpoint of interest anyway. We are currently developing this assay using fluorescently labeled blood protein-albumin.
Bibliography Type:	Description: (Last Updated: 03/04/2024)
Articles in Peer-reviewed Journals	Grabham P, Hu B, Sharma P, Geard C. "Effects of ionizing radiation on 3-Dimensional human vessel models: Differential effects according to radiation quality and cellular development." Radiation Research (in press), September 2010. , Sep-2010